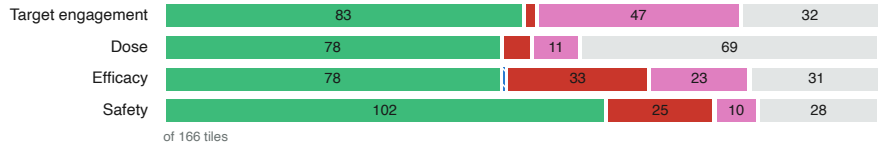
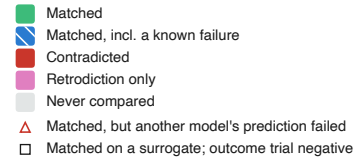
a





Panel b: every tile; diseases by global DALYs in 2021 (WHO Global Health Estimates), then those without a WHO cause by area.

b

		Target engagement	Dose	Efficacy	Safety
#1	**acute coronary syndrome** APOA1 replacement / gene therapy (3)	**2015** ApoE-/- mice infused with apoA-I Milano/phospholipid	**2017** CER-001 4-week repeat-dose animal toxicology (every second day)	**2024** Common APOA1 variants raising apoA-I (Copenhagen City… △ MDCO-216 no regression vs…	**2016** Nonclinical toxicology of CSL112 Model data unpublished
#2	**type 2 diabetes mellitus** GLP1R agonists (12)	**2011** Cynomolgus monkeys and r… Glucose-dependent insulin rose in humans	**2011** Rat and cynomolgus PK (dulaglutide)	**2022** GLP1R Gly168Ser genotype, GLP-1 infusion clamp in healthy volunteers	**2014** 1-year cynomolgus thyroid C-cell study (dulaglutide)
	type 2 diabetes mellitus GCK activators (9)	**2011** Rodent models of type 2 diabetes	**2016** Translational PK/PD model for sinogliatin (dorzagliatin), scaled from healthy-volunteer data and…	**2016** People with GCK-activating or GKRP loss-of-function variants △ AZD1656 missed 4-month HbA1c	**2022** Rodent models of type 2 diabetes Dorzagliatin raised hyperlipidaemia
	type 2 diabetes mellitus HSD11B1 inhibitors (8)	**2021** DIO mice, 11β-HSD1 inhibitor △ Adipose inhibition held 14 days	**2018** BMS-823778 PBPK, CYP2C19/UGT1A4 genotypes No checkable prediction	**2022** 11β-HSD1 knockout mouse Complete inhibition, no glucose lowering	**2022** Cortisone reductase deficiency: people with loss-of-function HSD11B1 or H6PD
	type 2 diabetes mellitus DPP4 inhibitors (7)	**2015** Rodent plasma DPP-4… 77-89% trough DPP-4 inhibition, GLP-1 x2	**2015** Dog and monkey pharmacokinetics scaled allometrically to a human half-life…	**2023** Population PK/PD plus… PK/PD-MBMA HbA1c forecast matched Phase 3	**2015** hERG channel assay plus dog and monkey ECG/telemetry (QTc)… △ Monkey skin harm not in patients
	type 2 diabetes mellitus INSR agonists (6)	**2016** STZ-diabetic rat single-dose PK/glucodynamics (Lilly) No checkable prediction	**2016** STZ-diabetic rat single-dose PK/glucodynamics (Lilly) No checkable prediction	**2022** STZ-diabetic rat single-dos… Efsitora HbA1c non-inferior to degludec	**2015** In vitro mitogenicity in L6-hIR, HMEC and Saos-2 cells △ Icodec: more level 2/3…
	type 2 diabetes mellitus GCGR/GLP1R agonists (5)	**2023** Human GLP-1 + glucagon co-infusion challenge (overweight volunteers without diabetes)	**2020** Diet-induced obese mouse GI intolerance capped dose below glucagon engagement	**2018** Human GLP-1 + glucagon co-infusion challenge (overweight volunteers without diabetes)	**2018** Diet-induced obese mouse △ Vomiting 17 % in patients
	type 2 diabetes mellitus SLC5A2 inhibitors (5)	**2012** Rodent hyperglycaemia models (90%-pancreatectomized r… Rat RTG drop reproduced in men	**2011** Rat PK allometric scaling (ertugliflozin)	**2021** People with SLC5A2… Genetic proxy reproduces trial direction	**2023** Rat 2-year carcinogenicity and chronic toxicology (SD rat, CD… △ Human bone loss, not rat…
	type 2 diabetes mellitus GCGR antagonists (4)	**2024** Glucagon challenge in… Cat glucagon challenge retrodicts human blockade	**2016** Glucagon-challenge hGCGR mice and rhesus monkeys; diabetic mice (small-molecule GRAs)	**2015** Gcgr-/- knockout mouse (genetic full antagonism)	**2016** Glucagon-challenge hGCG… LDL-C and ALT rises with the modelled compound
	type 2 diabetes mellitus GPR119 agonists (4)	**2014** GPR119-agonist mice, OGTT GLP-1/GIP/insulin flat in T2D; PYY rose	**2007** GPR119-agonist mice, OGTT No checkable prediction	**2018** GPR119-agonist mice, OGTT Firuglipel failed HbA1c; comparator worked	**2017** GPR119-agonist mice, OGTT
	type 2 diabetes mellitus CCR2 antagonists (3)	**2013** Diet-induced obese / db/db mice dosed with a CCR2 antagonist Not measured in patients	**2013** Human-CCR2 knock-in DIO / db/db mice dosed with CCX140-B No checkable prediction	**2013** Diet-induced obese / db/db mice dosed with a CCR2 antagonist	**2005** Diet-induced obese / db/db mice dosed with a CCR2 antagonist No checkable prediction
#3	**COPD** CHRM3 antagonists (6)	**2018** Human isolated bronchi:… Tissue offset >10 h, patients bronchodilated 24 h	**2016** Pfizer guinea-pig tracheal-strip relaxation (slow M3 off-rate) and in vivo…	**2018** Human isolated bronchi:… Trough FEV1 primary met, as predicted	**2016** Rat bronchoprotection and antisialagogue; dog bronchospasm △ Stopped; preclinical unexplained
	COPD ADRB2 agonists (4)	**2016** Guinea-pig, human bronchi,… Trough FEV1 up at all doses in COPD	**2016** Guinea-pig, human bronchi, dog bronchoprotection (abediterol)	**2016** ADRB2 coding-variant… ADRB2 genotype did not predict response	**2016** Guinea-pig, human bronchi, dog bronchoprotection (abediterol) △ Dose-dependent heart-rate rise…
	COPD MAPK14 inhibitors (4)	**2015** Rodent LPS lung neutrophilia	**2020** Acumapimod PBPK DDI model, CYP3A4/P-gp inhibitors	**2019** Mouse tobacco-smoke airway-inflammation model (SD-28… Acumapimod missed FEV1 primary	**2018** Human alveolar macrophage and whole-blood LPS-TNF potency with rat inhaled lung PK (AZD7624)
#4	**HIV** HIV RT (NNRTI) inhibitors (3)	**2015** HIV-1 culture vs wild-type a… Doravirine monotherapy cut HIV RNA 1.3 log10	**2020** In-silico doravirine PK model with rifabutin from the single-dose DDI, setting 100 mg twice daily	**2017** HIV-1 culture vs wild-type and NNRTI-resistant virus, resistance… Phase 3 non-inferior to darunavir/r	**2011** Rodent and monkey tox as filed for fosdevirine and doravirine Seizures no tox species predicted
	HIV HIV RT (NRTI) inhibitors (3)	**2013** HIV-1 replication in cell culture (CEM/PBMC/MT-4) with… Monotherapy cut HIV RNA ~1 log10	**2013** HIV-1 replication in cell culture (CEM/PBMC/MT-4) with… No checkable prediction	**2021** HIV-1 replication in cell cultu… Resistant-virus activity failed in patients	**2014** DNA polymerase-gamma / mtDNA-depletion assays in cell culture (CEM; later 19-day primary…
	HIV HIV integrase (INSTI) inhibitors (3)	**2011** HIV-1 replication in cell culture (MT-4/PBMC)	**2009** Protein-adjusted cell-culture potency as plasma trough target	**2012** In-vitro susceptibility of INSTI-resistant HIV-1 (site-directed mutant clones, fold-change)	**2022** Rat and rabbit embryo-fetal development studies △ Human NTD signal gone
	HIV HIV maturation (Gag) inhibitors (3)	**2021** HIV-1 replication in cell culture against patient-derived and site-directed Gag-polymorphic…	**2023** Population PK/Emax model of Phase 2a monotherapy viral load; simulation picked Phase 2b doses	**2023** HIV-1 replication in cell culture against patient-derived and site-directed Gag-polymorphic…	**2018** Nonclinical repeat-dose toxicology (rat and dog;… △ Animal gastric lesions absent in…
#5	**NSCLC** EGFR inhibitors (18)	**2017** T790M/sensitising-mutant NSCLC cells and mice	**2019** PC9 brain-metastasis mice; monkey osimertinib PET	**2023** Patient-derived lung cancer organoids △ EXCLAIM-2 fails primary PFS	**2021** hERG, rat and dog tox, dog telemetry QT prolongation the package cleared
	NSCLC ALK inhibitors (6)	**2019** Ba/F3 EML4-ALK L1196M cells and crizotinib-resistant patient-derived lines	**2018** EML4-ALK NSCLC xenografts (H2228/H3122, subcutaneous and intracranial)	**2019** Ba/F3 EML4-ALK L1196M cells and crizotinib-resistant patient-derived lines	**2018** Rat, monkey and dog toxicology
	NSCLC MET inhibitors (5)	**2019** METex14 and MET-amplified human tumour xenografts/PDX and HGF/MET allografts in mice	**2020** METex14 and MET-amplified human tumour xenografts/PDX and HGF/MET allografts in mice	**2023** Patient-derived lung cancer organoids, drug-sensitivity tested to steer the patient's next-line therapy	**2021** Mouse xenograft tolerability; GLP rat (26-week) and beagle dog… △ Oedema and ILD unpredicted
	NSCLC anti-IGF1R antibodies (4)	**2011** Tumour cells and xenografts given h7C10	**2005** A549 NSCLC xenografts in nude mice No checkable prediction	**2011** A549 NSCLC xenografts in… Dalotuzumab + erlotinib: no PFS gain	**2005** A549 NSCLC xenografts in nude mice No checkable prediction
	NSCLC anti-PDCD1 antibodies (4)	**2020** Cynomolgus pharmacology and repeat-dose tox, PD-1 receptor occupancy (toripalimab, dostarlimab)	**2023** Cynomolgus pharmacology and repeat-dose tox, PD-1 receptor occupancy (toripalimab, dostarlimab)	**2023** MC38 tumours in NSG mice engrafted with allogeneic human CD8+ T cells, dosed with toripalimab	**2022** PD-1 knockout mice △ Monkeys silent; patients had irAEs
	NSCLC pan-HER inhibitors (4)	**2011** EGFR del19/L858R NSCLC lines and HCC827 xenografts	**2021** EGFR/HER2 exon 20 insertion Ba/F3 lines, PDX and GEMM… Not measured in patients	**2017** EGFR del19/L858R NSCLC lines and HCC827 xenografts △ 0/11 exon-20 responses	**2020** Exon 20/ERBB2-mutant patient lines, xenografts (tarloxotinib) △ PDX silent on grade 3 diarrhoea
	NSCLC HSP90AA1/HSP90AB1 inhibitors (3)	**2013** NSCLC lines and xenografts No checkable prediction	**2013** NSCLC lines and xenografts	**2013** Kras G12D; p53-deficient autochthonous lung adenocarcino… △ Docetaxel phase 3 failed, HR 1.11	**2015** Beagle dog repeat-dose retinal toxicity model (ERG, pupilla… △ No visual harm on ganetespib in…
	NSCLC VEGFR multikinase inhibitors (3)	**2022** Mouse Lewis lung carcinoma, VEGFR TKI + RT + anti-PD-1 Not measured in patients	**2022** Mouse Lewis lung carcinoma, VEGFR TKI + RT + anti-PD-1 No checkable prediction	**2016** RET-fusion lung, thyroid and Ba/F3 cells; NIH3T3 xenografts △ Post-LEAP lenvatinib model…	**2019** GLP repeat-dose toxicology in rat plus monkey (and dog)
	NSCLC anti-CTLA4 antibodies (3)	**2010** B16, MC38 and CT26 mouse tumours, anti-CTLA-4 plus anti-PD-1 Not measured in patients	**2018** B16, MC38 and CT26 mouse tumours, anti-CTLA-4 plus anti-PD-1 No checkable prediction	**2020** B16, MC38 and CT26 mous… Ipilimumab added nothing to anti-PD-1	**2010** CTLA-4 knockout mice △ Monkeys missed fatal immune tox
#6	**migraine disorder** CGRP receptor antagonists (4)	**2020** Rhesus capsaicin-induced dermal vasodilation (CIDV) Rhesus EC50 3.2 vs human 2.6 nM	**2014** Rhesus CIDV PK/PD target Predicted 20 mg; needed 200 mg	**2020** Rat sumatriptan-primed… Rat allodynia reproduces ubrogepant efficacy	**2026** Mouse MCAO stroke, gepant pretreatment △ Animal tox missed MK-3207 liver
	migraine disorder anti-CGRP ligand antibodies (3)	**2018** Human ex-vivo cerebral, middle meningeal and abdominal… Not measured in patients	**2008** Rat and cynomolgus evoked neurogenic vasodilation (electrical… No checkable prediction	**2014** Rat and cynomolgus evoked neurogenic vasodilation (electrical… △ Pediatric CM Phase 3 primary…	**2025** Rat blood pressure, heart rate on anti-CGRP mAb; monkey safety Postmarketing hypertension warning
#7	**Alzheimer disease** BACE1 inhibitors (4)	**2015** PDAPP mice and cannulated beagle dogs (brain/CSF Aβ) Human CSF Aβ fell as in dogs	**2015** Dog CSF Aβ PK/PD scaled to humans No checkable prediction	**2021** PDAPP (APP V717F) transgenic mice No change in tau PET	**2018** Mice chronically dosed with BACE1 inhibitors △ Liver injury missed by tox